

Eric Gitangu

Senior Software Engineer | Distributed Systems, Platform Architecture & Security

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PROFESSIONAL SUMMARY

Software engineer with 11 years in production systems across fintech, healthcare, energy access, logistics and application security. I currently own architecture and security for a multi-surface payments and mobility platform, a Mastercard partner, where I removed an authentication bypass, closed money-correctness defects, mapped the platform to payments compliance requirements, and built card and mobile money rails within PCI DSS scope. Before that I led platform and quality engineering for a pay-as-you-go solar company in seven African countries, founded an ML security company for medical device software, and led the team behind a digital addressing pilot with the Nairobi County Government. I design event-driven systems, run the cloud and edge they live on (AWS, GCP, Azure, Cloudflare; Terraform and Ansible; nginx, load balancing, rate limiting, high availability), plan for failure, instrument what I ship, and lead teams through review, mentoring and clear technical direction. TypeScript, Java, PHP, Python, Rust, Go and C#.

EXPERIENCE

Senior Full Stack Engineer, Architecture and Security, Moovn Technologies (a Mastercard partner)

Feb 2026 – Present

Nairobi, Kenya, remote

Own delivery, security and architecture for a multi-surface fintech mobility platform: consumer, partner and admin web apps and a mobile API over a service backend, covering money movement, identity and regulated payments compliance.

- Removed a critical authentication bypass and migrated stored credentials to modern hashing, verified on the live path.
- Closed money-correctness defects in locality routing and currency precision across the financial surfaces, and added idempotency that fails closed on every money path.
- Built the card top-up rail on Mastercard Payment Gateway Services hosted sessions, so card data never reaches our servers, and the M-Pesa, MTN and Tigo mobile money rails with per-country routing. Mapped controls to PCI DSS SAQ-A.
- Designed the release process: immutable releases, symlink activation, a post-deploy smoke that rolls back automatically, and a release guard on money-path files. Deploy and rollback are each one command.
- Designed a cross-product trust score engine with signed partner event ingestion, a tamper-evident ledger and a nightly recompute, rolled out behind feature flags to the admin console, rider portal and mobile API.
- Hardened access and the SDLC: key-only SSH, MFA for admins, audit trails, secret scanning and test gates in git hooks, pentest remediation, encrypted off-site backups with restore drills.
- Rebuilt three front-end surfaces in typed, tested React, documented the API surface, and author the architecture decision records that name trade-offs and failure modes.

PHP TypeScript Next.js React Node.js MongoDB Redis Nginx Prometheus Grafana

Team Lead, Platform and Quality Engineering, ENGIE Energy Access, later Ignite Energy Access

May 2025 – Jan 2026

Nairobi, Kenya, hybrid

Pay-as-you-go solar platform serving seven African markets with millions of daily transactions.

- Designed the event-driven payment architecture: RabbitMQ-backed bounded contexts for payment, token and subscription that decoupled the synchronous PAYG lifecycle and removed peak-load timeouts across all seven markets.
- Instrumented tracing and monitoring across Ruby and Node.js services (OpenTelemetry, Prometheus, Grafana, Jaeger, ELK) with SLOs and error budgets; detection of payment anomalies fell from hours to under five minutes. Built the Redshift and Kafka telemetry and payment reporting pipeline.
- Led the shift-left quality programme: Playwright and Cypress suites covering 900 critical-path scenarios across payments, tokens, IoT provisioning and subscriptions. Regression fell from three days to 45 minutes; releases rose from 2 to 12 a week.
- Led the Rails 5 to 7.2 upgrade underpinning payment processing, PAYG token dispatch, IoT diagnostics and firmware updates, including a staged rollout and an incident rollback with no customer-facing downtime.
- Line-managed four engineers across Kenya, Uganda and Tanzania; two were promoted to senior during my tenure.

Ruby on Rails Node.js RabbitMQ Kafka Redshift PostgreSQL Playwright Cypress OpenTelemetry Kubernetes

Director of Engineering, Vishnu Systems, Inc.

Jan 2024 – Aug 2025

Atlanta, GA, remote

Founded and shipped an ML security and robustness platform for software in medical devices (SiMD and SaMD), where model failure has patient-safety implications. Ran concurrently with Baw Bab.

- Architected the backend: FastAPI orchestration, a Rust and PyO3 evaluation core, Go middleware for rate limiting and worker coordination, and GPU evaluation workers on AWS with idempotent workers and dead-letter queues.
- Shipped ML pipelines combining CNNs, retrieval-augmented generation and graph neural networks to detect adversarial attacks on medical ML models, orchestrated with Airflow and tracked with MLflow.
- Built compliance tooling that maps model evaluation results to FDA, HIPAA and IEC 62304 requirements, with LLM-generated regulatory documentation, in a category with few established vendors.
- Operated the production stack (SQS, SNS, Kinesis, hosted Kafka) and carried on-call for customer-facing evaluation SLAs.

Python FastAPI Rust PyO3 Go AWS Kafka Airflow MLflow PyTorch

Project Lead Developer, Baw Bab Technologies, Inc.

Apr 2023 – Aug 2025

Seattle, WA and Nairobi, remote

Digital addressing infrastructure for last-mile delivery in African urban areas without formal addresses. Ran concurrently with Vishnu Systems.

- Led a five-engineer team and shipped the Nairobi County Government pilot serving 2.3 million residents, the second digital addressing programme in Africa, integrated with the county land registry.
- Designed the computer vision and GIS pipeline over Azure, Google and OpenStreetMap satellite imagery that converts residential structures to Google Plus Codes, tuned for low-bandwidth field use.
- Owned the stack: Node.js APIs, a mobile-first React front end for field agents, Google Cloud Vision and GIS services on a Pub/Sub event bus, MongoDB and Kafka for spatial data, deployed on GCP Cloud Run and Azure.

Node.js React Kafka MongoDB PostGIS Google Cloud Vision GCP Cloud Run Azure

Software Engineer (open source, seasonal), TruStacks

Jan 2023 – Aug 2025

Remote

- Contributed pipelines to TruStacks, an open-source DevOps orchestration platform of declarative delivery stacks: dependency fetching, tests, builds, staging and security scanning across frameworks.

Go GitHub Actions Docker Kubernetes

Career break, Health and professional development

Jul 2022 – Apr 2023

Atlanta, GA

- Time taken for health and wellbeing and for structured professional development, including certifications in cloud, AI and security.

Senior Software Engineer, Full Stack (contract), Reinsurance Group of America

Jan 2022 – Jul 2022

United States, remote

Small team delivering internal applications for reinsurance operations across several countries, each localised with country-specific roles and access.

- Delivered features test-first with Jest, and end-to-end coverage with Cucumber, Puppeteer and Cypress, on a hybrid Scrum and Kanban cycle with after-hours support duty.
- Worked across the MERN and MEAN stacks (React, Angular, Node.js, MongoDB) and the legacy Grails and Groovy services with PostgreSQL.

React Angular Node.js MongoDB PostgreSQL Grails Jest Cypress

Lead Software Engineer (contract), Wagelyft

Jan 2019 – Feb 2022

Nairobi, Kenya, remote

Fintech and e-commerce SaaS for small businesses in Kenya, delivered by a freelance team I led and mentored.

- Built C2B and B2B cloud products on Reactive Spring with Angular and on the MERN and MEAN stacks, on AWS with Docker and Kubernetes, with M-Pesa payments and multi-currency support.
- Ran the mentorship programme that grew the team: 15 developers mentored through code review, system design and career coaching, six of whom moved to mid or senior roles.

Java Spring Angular React Node.js MongoDB AWS Kubernetes

Software Engineer, Vecima Networks Inc.

Jan 2018 – Jan 2019

Peachtree City, GA

- Wrote C and embedded assembly for petabyte-scale video streaming pipelines with deterministic store-and-replay and zero frame loss, and designed the redundant data paths and self-healing recovery for content delivery.

C Assembly Linux CDN

Software Engineer, Veracode

Dec 2015 – Feb 2018

Burlington, MA

Application security platform for static, dynamic and composition analysis of customer code.

- Built and extended the customer integrations, including the Jira and Visual Studio plugins and the Jenkins and GitLab CI integrations, so customers could run scans and receive findings inside their own pipelines.
- Worked on the core platform in Java (Spring MVC, Spring Boot, Hibernate), C# .NET and Python on Docker, Kubernetes and AWS, including the TLS 1.2 enforcement upgrade for regulated customers.

Java Spring Boot Hibernate C#/.NET Python Jenkins Docker AWS

Software Engineer, Intern, IBM

Sep 2013 – May 2014

Littleton, MA

- Built the Selenium test automation for IBM Web Experience Factory with the principal architect and QA lead, integrated it into the Jenkins pipelines with Python and Bash, and built the reporting that separated real failures from false positives.

Java Selenium Jenkins Python

SELECTED PROJECTS

PesaCore, a .NET 10 core banking platform · [pesacore-web-261341881894.africa-south1.run.app](#)

Core banking platform on ASP.NET Core 10 with a Blazor front end: CQRS with idempotent transfers, PostgreSQL and Redis, OpenTelemetry, a hardened nginx edge, deployed on GCP Cloud Run.

C# .NET 10 ASP.NET Core Clean Architecture MediatR CQRS Polly FluentValidation Entity Framework Core AutoMapper

LendStream · [nanolend-app.vercel.app](#)

Micro-lending reference platform: Java 21 Spring Boot and Clojure services on Kafka with CQRS, event sourcing and a transactional outbox.

Java 21 Spring Boot Clojure React TypeScript Kafka QStash PostgreSQL Redis Docker

UniCorns, multi-tenant SaaS · [unicorns.run](#)

Multi-tenant SaaS for clinics, pharmacies and retail in Kenya: eight Rust services on AWS Lambda, tenant isolation on Cognito claims, offline-first web and mobile clients, M-Pesa, Airtel, Stripe and SHA claims rails.

Rust AWS Lambda DynamoDB Next.js Kotlin Claude AI M-Pesa

Vunachain, agricultural traceability

Supply chain traceability for tea and coffee exports under EU deforestation rules: Django and PostGIS geospatial checks, harvest records anchored on Celo, M-Pesa payouts. Demo stage.

Rust AWS Lambda ARM64/Graviton EventBridge SQS DLQs S3 RDS PostgreSQL CloudWatch X-Ray

SwarmBench, multi-agent LLM study · [developer.ericgitangu.com/pdf/research-paper.pdf](#)

Empirical study of single-agent versus multi-agent LLM performance on software tasks, written as a Harbor benchmark across codebases up to 303K lines. No multi-agent advantage found for tool-using agents.

Kimi K2.5 Fireworks AI Harbor Docker Python pandoc WeasyPrint LLM-as-judge Multi-Agent Systems Benchmark Design

AgentPlane, AI project planning · [agent-plane.ericgitangu.com](#)

Project management tool where a three-agent LangGraph pipeline breaks an epic into estimated sub-tasks with a replayable audit log. FastAPI and Next.js with cost-bounded LLM calls.

Python 3.12 FastAPI LangGraph LangChain Anthropic Next.js 14 TypeScript TanStack Query CockroachDB MongoDB Atlas

PawaCloud, AI cloud advisor · [pawacloud-web.vercel.app](#)

Conversational cloud-architecture advisor on Gemini 2.5 Flash with a Rust and PyO3 text-processing core, server-sent event streaming, Google OAuth and PostgreSQL, deployed on GCP Cloud Run.

Python FastAPI Rust PyO3 Gemini AI Next.js TailwindCSS PostgreSQL Google Cloud Cloud Run

CORE TECHNICAL SKILLS

Architecture & Systems

Distributed systems · Microservices · Event-driven architecture · CQRS · Event sourcing · Saga orchestration · Hexagonal / ports and adapters · Domain-driven design · Backend-for-frontend · Multi-tenant SaaS · API design (REST, GraphQL, gRPC, OpenAPI) · Resiliency patterns (circuit breaker, back-off, idempotency) · Strangler-fig migration · Zero-trust auth · Scale-to-zero serverless design · Capacity and failure-mode analysis

Payments, Security & Compliance

PCI DSS (SAQ-A) · Tokenization / hosted payment sessions · Idempotency & fail-closed money paths · Settlement & reconciliation · OAuth2/JWT · OWASP Top 10 · Secure SDLC / threat modelling · MFA (WebAuthn / TOTP) · Audit hash-chains · NIST SSDF · HIPAA · FHIR/HL7 · AES-256-GCM

Cloud & DevOps

AWS · GCP · Azure · Cloudflare (Workers, DNS, CDN, WAF) · Nginx (edge reverse proxy) · Vercel · Fly.io · Kubernetes · Docker · Terraform · Ansible · CloudFormation / CDK · GitHub Actions · GitLab CI · Jenkins · Atomic releases and rollback

Edge (Cloudflare, Nginx), Networking & Reliability

Nginx reverse proxy and API gateway · Load balancing · High availability and failover · Rate limiting · TLS 1.2/1.3, HSTS, CSP · ModSecurity / OWASP CRS · fail2ban and SSH hardening · Prometheus · Grafana · Loki · OpenTelemetry · Jaeger · ELK · SLOs and error budgets · Chaos and load testing (K6, Locust) · Linux (Ubuntu, Debian)

Backend & APIs

TypeScript · Node.js · Java · Spring Boot · PHP · Python · Rust · Go · C#/.NET · Kafka · Scala · Clojure · C/C++ · Django · FastAPI · Groovy/Grails

AI/ML

TensorFlow · PyTorch · LLMs · OpenAI · Anthropic Claude · Google Gemini · RAG Systems · CNNs · GNNs · Transformers · Apache Airflow · MLflow · Vertex AI · pgvector

Databases & Data

PostgreSQL · MongoDB · Redis · DynamoDB · MySQL · Kafka · RabbitMQ · Redshift · ETL/ELT Pipelines

Frontend & Mobile

React · Next.js · TypeScript · Angular · Vue.js · Tailwind CSS · PWA · Kotlin + Jetpack Compose · Swift/SwiftUI · Electron

EDUCATION, CERTIFICATIONS AND OPEN SOURCE

Bachelor of Science in Computer Science, minor in Mathematics, University of Massachusetts at Lowell

Sep 2011 – May 2015

- Microsoft Scholarship, 2012 to 2013, for academic excellence in software engineering
- Google Scholarship, 2011 to 2012, for achievement in computer science coursework and research
- Dean's List, nine semesters, 2010 to 2015
- Omicron Delta Kappa National Leadership Honor Society
- **Open source and certifications:** Public GitHub repositories in Rust, Go, TypeScript, Python, Java and C#, plus contributions to TruStacks. 80 certifications in AWS, Azure, GCP, AI, blockchain and application security.
- **Mentorship and teaching:** 15 engineers mentored since 2019, six promoted to senior or lead roles. Volunteer instructor for a CodePath.org advanced algorithms cohort in 2024.